

Magnetic Field Scanner - Bill of Materials (BOM)

Core Electronics

Part Description	Quantity	Link / Vendor	Notes
Waveshare ESP32-S3-Touch-LCD-3.49 (v3 / PCBA 1.1)	1	[Link Placeholder]	CRITICAL INTEGRATED PLATFORM: This is not just a display. It contains the ESP32-S3 MCU, QMI8658 IMU, TCA9554 I/O Expander, and Audio Codec physically built into the board. Do not buy the IMU or Expander separately.
PNI RM3100 Magneto-Inductive Sensor Breakout	2	[Link Placeholder]	One for the TIP (0") and one for the REF (24"). An optional 3rd sensor can be placed at the NEAR position (8").
Micro-SD Card (SPI)	1	[Link Placeholder]	Used for high-speed logging.

Peripheral Components

Part Description	Quantity	Link / Vendor	Notes
2W 8 Ohm 1" Speaker	1	[Link Placeholder]	Driven by the Waveshare's integrated audio codec via I2S.
4.7K Ohm Resistor	2	[Link Placeholder]	I2C pull-up resistors (if not utilizing the internal pull-ups).
CAT6 Ethernet Cabling	[Qty Placeholder]	[Link Placeholder]	Used for highly-shielded data and power transmission to the remote RM3100 sensors. Separate and different from the I2C bus wiring.

Mechanical & Jig Components

Part Description	Quantity	Link / Vendor	Notes
Fiberglass Tube (1" OD, 3/4" ID, 46.5" Long)	[Qty Placeholder]	[Link Placeholder]	Acts as the rigid non-magnetic shaft of the wand.
Fiberglass Rod (1/4" OD, 48" Long)	[Qty Placeholder]	[Link Placeholder]	Internal structural reinforcement or wire-routing guide.

Part Description	Quantity	Link / Vendor	Notes
3D Printed Parts (.stl)	[Qty Placeholder]	Local 3D Printer	Board_Holder_REF&NEAR_Part1_(1.7 Board_Holder_REF&NEAR_Part2_(0.7 etc. found in docs/mechanical/stl.

Characterization Setup (Optional)

Part Description	Quantity	Link / Vendor	Notes
5/8" x 36" Steel Rebar	1	Home Depot / Local	Used as a standard magnetic monopole target for open-air distance tests.
Plywood Sheet & Wooden Guides	1	Home Depot / Local	Used to build a magnetically-inert sliding jig for precision measurement testing.